LAPORAN WORKSHOP DEVELOPER OPERATIONAL

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E41230057

PROGRAM STUDI TEKNIK INFORMATIKA JURUSAN

TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

2026

1. Insttal minikube

```
PS C:\Users\user> winget install Kubernetes.minikube
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to
rly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes  [N] No: y
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernete
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party pa
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minik
  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
```

```
PS C:\Users\user> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 10 Pro 22H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 5.43 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 8.23 Mi
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

```
PS C:\Users\user> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   91s   v1.35.1
```

```
PS C:\Users\user> mkdir kubernetes


    Directory: C:\Users\user


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/29/2026   8:48 AM                kubernetes


PS C:\Users\user> cd kubernetes
PS C:\Users\user\kubernetes> code .
```

2. Membuat directori Kubernetes dan membuat file deployment.yaml dan service.ymal

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

```
PS C:\Users\user\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
PS C:\Users\user\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service created
```

3. Pastikan semua server berjalan lalu akses Alamat IP

```
PS C:\Users\user\kubernetes> kubectl get all
NAME                              READY   STATUS    RESTARTS   AGE
pod/web-nginx-748d764f7-mpx7j     1/1     Running   0          57s
pod/web-nginx-748d764f7-pstbs     1/1     Running   0          57s
pod/web-nginx-748d764f7-qrq4p     1/1     Running   0          57s

NAME                          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)
     AGE
service/kubernetes            ClusterIP   10.96.0.1        <none>        443/TCP
     7m23s
service/web-nginx-service     NodePort    10.110.102.174   <none>        80:32251/T
CP   36s

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     3/3     3            3           57s

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         3       57s
```

```
PS C:\Users\user\kubernetes> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-mpx7j     1/1     Running   0          101s
web-nginx-748d764f7-pstbs     1/1     Running   0          101s
web-nginx-748d764f7-qrq4p     1/1     Running   0          101s
```

```
PS C:\Users\user\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-mpx7j     1/1     Running   0          2m13s
web-nginx-748d764f7-pstbs     1/1     Running   0          2m13s
web-nginx-748d764f7-qrq4p     1/1     Running   0          2m13s
```

Welcome to nginx!

127.0.0.1:52929

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

```
PS C:\Users\user\kubernetes> minikube service web-nginx-service
┌───────────┬───────────────────┬─────────────┬──────────────────────────┐
│ NAMESPACE │       NAME        │ TARGET PORT │           URL            │
├───────────┼───────────────────┼─────────────┼──────────────────────────┤
│ default   │ web-nginx-service │ 80          │ http://192.168.49.2:32251 │
└───────────┴───────────────────┴─────────────┴──────────────────────────┘
🔗  Starting tunnel for service web-nginx-service.
┌───────────┬───────────────────┬─────────────┬──────────────────────────
│ NAMESPACE │       NAME        │ TARGET PORT │          URL
```

Yang terjadi di terminal 1 jika, terminal 2 dan 3 di jalankan maka dapat di lihat dari terminal 1 nya

```
PS C:\Users\user\kubernetes> kubectl get pods -w
NAME                        READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-mpx7j   1/1     Running   0          8m32s
web-nginx-748d764f7-pstbs   1/1     Running   0          8m32s
web-nginx-748d764f7-qrq4p   1/1     Running   0          8m32s
web-nginx-748d764f7-24q5r   0/1     Pending   0          0s
web-nginx-748d764f7-6bg5w   0/1     Pending   0          0s
web-nginx-748d764f7-24q5r   0/1     Pending   0          0s
web-nginx-748d764f7-6bg5w   0/1     Pending   0          1s
web-nginx-748d764f7-24q5r   0/1     ContainerCreating   0
  1s
web-nginx-748d764f7-6bg5w   0/1     ContainerCreating   0
  1s
web-nginx-748d764f7-24q5r   1/1     Running             0
  11s
web-nginx-748d764f7-6bg5w   1/1     Running             0
  12s
web-nginx-748d764f7-24q5r   1/1     Terminating         0
  42s
web-nginx-748d764f7-6bg5w   1/1     Terminating         0
  42s
web-nginx-748d764f7-24q5r   1/1     Terminating         0
  42s
web-nginx-748d764f7-6bg5w   1/1     Terminating         0
  42s
web-nginx-748d764f7-6bg5w   0/1     Completed           0
  47s
web-nginx-748d764f7-24q5r   0/1     Completed           0
  47s
web-nginx-748d764f7-6bg5w   0/1     Completed           0
```

Terminal 2

```
PS C:\Users\user\kubernetes> kubectl scale deployment web-nginx
-replicas=5
deployment.apps/web-nginx scaled
```

Terminal 3

```
PS C:\Users\user\kubernetes> kubectl scale deployment web-nginx -
-replicas=3
deployment.apps/web-nginx scaled
```

Terminal lain

```
PS C:\Users\user\kubernetes> kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   3/3     3            3           12m
```

4. Ini jika ingin menambahkan node

```
PS C:\Users\user\kubernetes> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it migh
t cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
```

```
PS C:\Users\user\kubernetes> kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   23m   v1.35.1
minikube-m02   NotReady   <none>          52s   v1.35.1
PS C:\Users\user\kubernetes> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running
```

5. Jika ingin mendelet maka lakukan ini

```
PS C:\Users\user\kubernetes> kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
PS C:\Users\user\kubernetes> kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
PS C:\Users\user\kubernetes> minikube delete
🔥  Deleting "minikube" in docker ...
🔥  Deleting container "minikube" ...
🔥  Deleting container "minikube-m02" ...
🔥  Removing C:\Users\user\.minikube\machines\minikube ...
🔥  Removing C:\Users\user\.minikube\machines\minikube-m02 ...
💀  Removed all traces of the "minikube" cluster.
```

```
PS C:\Users\user\kubernetes> minikube start --driver=docker
😄  minikube v1.38.1 on Microsoft Windows 10 Pro 22H2
✨  Using the docker driver based on user configuration
❗  Starting v1.39.0, minikube will default to "containerd" conta
iner runtime. See #21973 for more info.
📌  Using Docker Desktop driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube"
cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: default-storageclass, storage-provisioner
🏄  Done! kubectl is now configured to use "minikube" cluster and
 "default" namespace by default
```